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CART 263 - Section A

Artist Statement II

Remastered 3D Snake Game

Presented to: Sabine Rosenberg

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The game's developers, Ashmitha Kanagiah, Ya Xuan Pang, and Nerly Cadet, worked on various aspects of the project before assembling it into a single, large game. We chose to go with the same theme—a snake game—for our second project. The primary goal was to take inspiration from our initial project and transform it into something more visually stunning, offering a more challenging gameplay. This time, we made the decision to use 3D materials in order to produce a different version of our previous project. The instructions screen appears when you launch the game. The initial page's layout was mostly influenced by arcade games, which frequently employ neon colors on top of dark backgrounds. The instructions describe how the game operates, including how to win and how to prevent losing. Additionally, it informed players about the game's interactivity: "Your microphone controls targets. The louder, the more targets appear." "Click and drag your mouse to rotate the camera." The screen that shows up after selecting “PLAY” is the game itself. On top of a 3D grid lay a shining purple sphere representing the target, and a shining soft pink cube representing the snake. To show the players how close they are to winning, the score is displayed on the upper right corner of the screen. As stated in the instructions, by turning on their microphones, players can interact with the game’s scale of difficulty. Moreover, they can play with the point of view of their choice by navigating through the canvas’ X,Y, and Z position by using the mouse. We also added stars in the background of our game to make it seem less boring. The player gets a starry night view instead of staring into a boring dark void. In the end, the player has the choice to play again after winning or losing the game.

Before agreeing to begin working on an enhanced version of our previous snake game, we first went over our past document, in which we explained our thoughts for the game’s design and content. Our original plan – both for our previous project and this one – was to make a magical girl game in which we would play the roles of the magical girls. The design of the game would imitate the one of a typical side-scrolling infinite flyer arcade game. However, in response to the critics we received for our prior project, we made the decision to attempt making the gameplay and aesthetic less basic and simple. To add to the game, sounds – determining the growth, the win or the death of the snakes –and textures we used.

The goal of the game is to make the snake as large as possible, with a maximum size of 20 points. In order to accomplish this, the snake must avoid touching itself or the sides of the grid, since doing so would result in the player’s loss. To determine whether the snake has collided with itself, the code calls a function named “checkSelfCollision();” The segments of the snake that are inaccessible are represented by a segment array in the object. The if statement then compares each of the head’s coordinates (X,Y, and Z). The function ends and reports a collision as soon as the if statement becomes true; if the loop ends without finding a match, no collision

has been recorded. Both the true and false are examined in the return. The “resetPosition()” function, which determines whether the snake has touched the grid’s edges, and the “checkCollision(target)” function, which determines whether the snake has contacted the target, follow a similar code. Both the snake and the target are classes, in which constructors were used to set the scene and initialize the state of both classes.

Additionally, we had a very collaborative teamwork. We decided to improve on our first project and kept adding new ideas as they came. Each member was assigned a different part and would send it in the group chat we created to be combined into one big master file. Yaxuan kept the master file again and made the commits on github. The game can be found on her repository. We communicated with each other, followed up with each other’s progress and sent all the files including the master file every time a new element was added for verification from the other teammates. Every member did their work on their own time again which caused some conflict within the team. Although we did keep constant communication, we started to have some problems with our coordination and timing. Due to the added pressure of other assignments piling up, our schedules were all over the place and resentment started to build up within the team. Despite this challenge, we were able to finish our work and resolve most of our personal conflicts. Overall, despite our teamwork not working out as great as last time, we completed a project that we were proud of, proving that our team is as strong as ever and that we are able to resolve our disagreements without affecting the quality of our work. As for the technical challenges we faced while making this game, our main struggle was actually making the game 3D. We struggled to make different classes for our game and ultimately decided to do everything in one javaScript file due to the uncertainty and risks it carried, assuming that three.js has a different way of making different classes. We tried following examples online but failed multiple times to make it work. Another struggle is working with the different lights and textures. We wanted the textures to appear seamless but the lighting would brighten it or make the colors look different and weird. After multiple attempts, we were able to match the lighting and how it hits on the textures as we preferred.

Furthermore, remastering our project into a 3D version was an interesting process. We already had a blueprint of how we wanted our game to look as we drew most of our inspiration from our game that we had given for the first project. For the functionalities, rules, and aesthetics, we were deeply inspired by our first version, which was in 2D. A lot of the game technicalities are the same as our first game; we simply tweaked and changed minor elements. All we needed to focus on was turning the entire gameplay into a 3D format. We did some research on other 3D gameplay to familiarize ourselves with the environment as well as to see if there were a certain way to work with it, since we are working with three dimensions rather than two. The article published by Medium was a major inspiration regarding the visual interface of our game. We

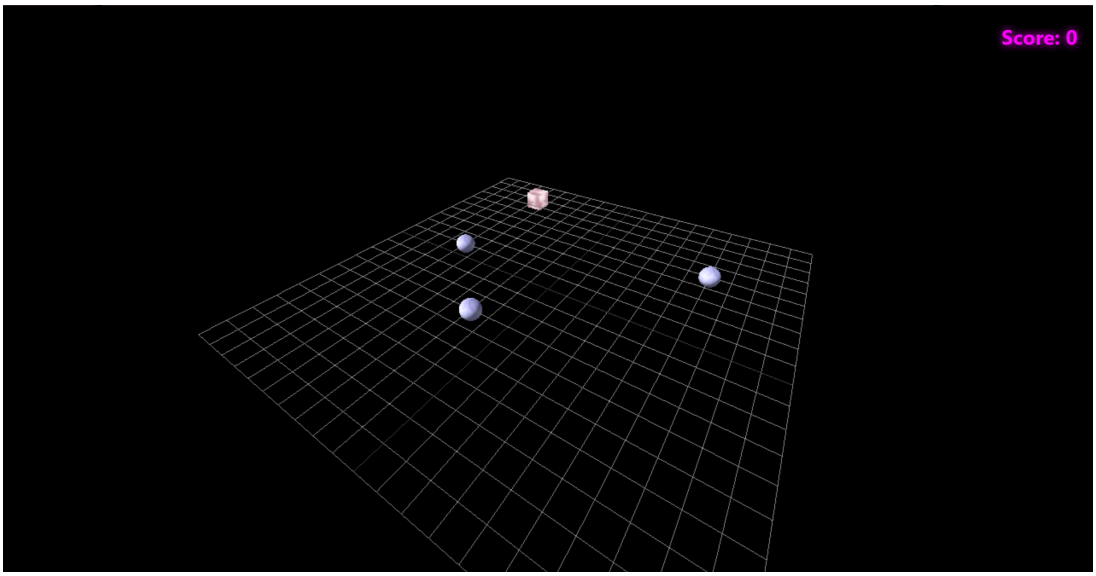
decided to go along with the same grid layout and background color as the one from the game *Hyperspeed*, made by the web developer Mina Pecheux. Having a simple white grid, where our game is being played, helped to put in perspective the 3D shapes on top of it. The black background also emphasizes the game and the colors of our game. Another resource that we came across is the research paper by Weiquan Mao and Jiyao Yuan from Stanford University titled 3D Snake Game. This research is the inspiration behind why we wanted to use orbit control, to make sure the user is aware that we can move the grid around and travel all around the 2D and 3D environments. Having different visualizations, such as from the top view, the bottom view and the side views, allows the player to see the game clearly as well as to immerse themselves in an interesting experience. It adds more depth to our game as well as making it feel more fun as we move away from the common two-dimensional layouts from most games. It also made us fascinated by how advanced we have become with technology and visuals, as we started with simple 2D games to now full-on 3D video games with such amazing and powerful graphics and models. Lastly, we used the *Texture Labs* online resource to search and find the textures that we wanted our snake and targets to be. It is an interesting resource with lots of free imagery and textures that we can use, which we definitely kept in mind for any future references or projects. Our research and hunt for inspiration were both helpful and insightful regarding the evolution of games and the game industry, as well as getting to go away with great features and ideas to incorporate into our own game for the project.

In conclusion, this project represents a successful balance between creative ambition and technical execution, specifically through our transition from a 2D foundation to a complex 3D environment. We explored more complex interactions such as microphone reactive gameplay. Our research, from arcade-inspired neon aesthetics to academic studies on 3D navigation, allowed us to build a project that is both theoretically grounded and visually immersive. The technical hurdles we faced, such as refining 3D lighting and managing seamless textures, served as learning moments that sharpened our problem-solving skills without compromising our artistic vision. While the pressures of the term tested our coordination and our ability to resolve internal conflicts reinforced our collaborative resilience. Ultimately, this version of the snake game stands as a testament to our growth as developers and our ability to research, adapt, and execute high-quality work as a cohesive team.

Link to game: https://yaxuanpang.github.io/cart263/topics/project2/

Link to the video: snakeGame.mp4

SCREEN GRAB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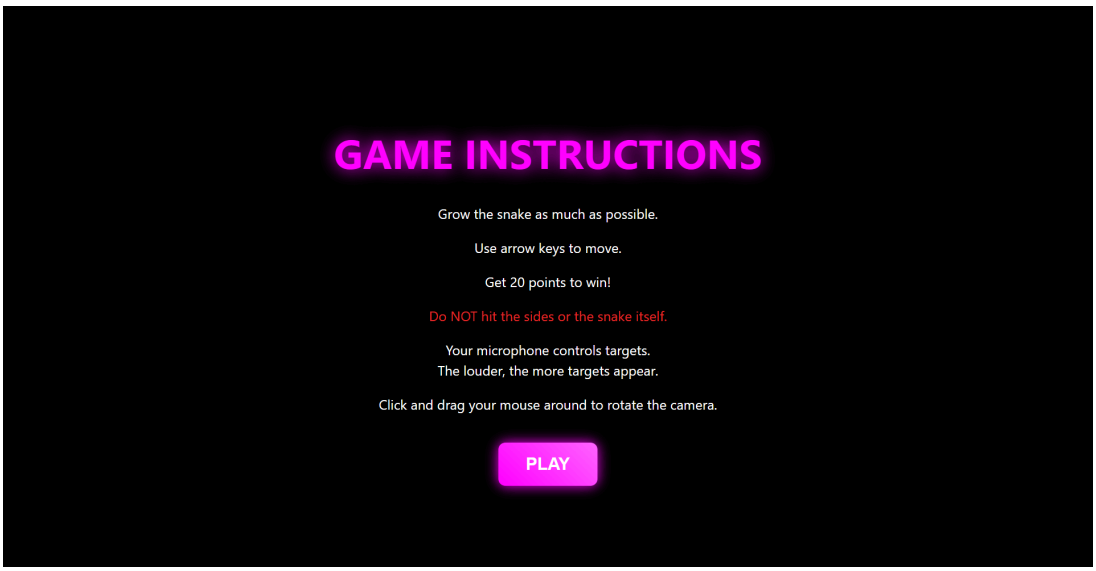
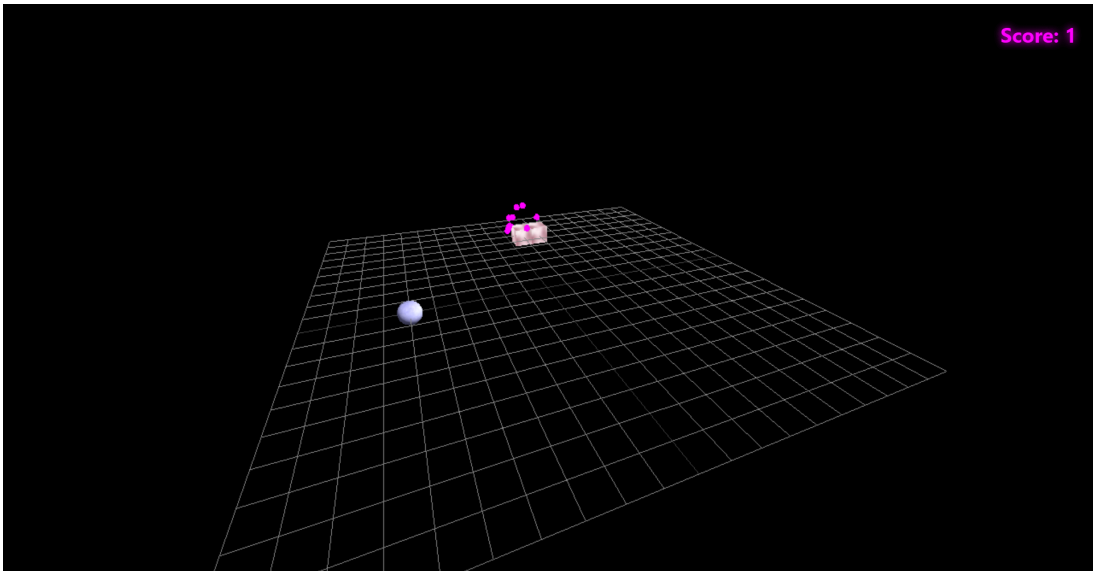
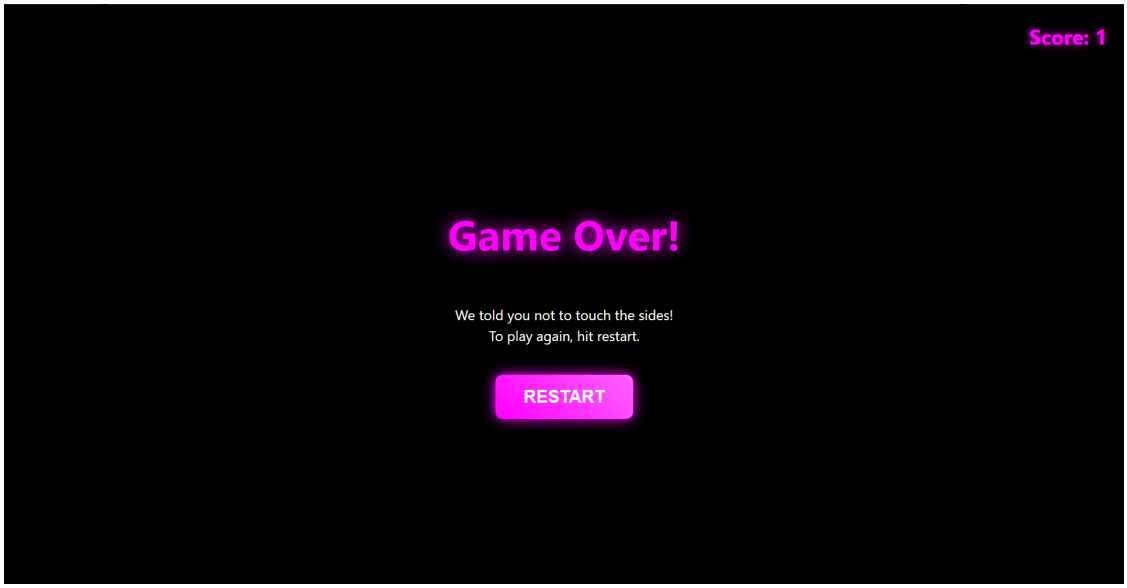
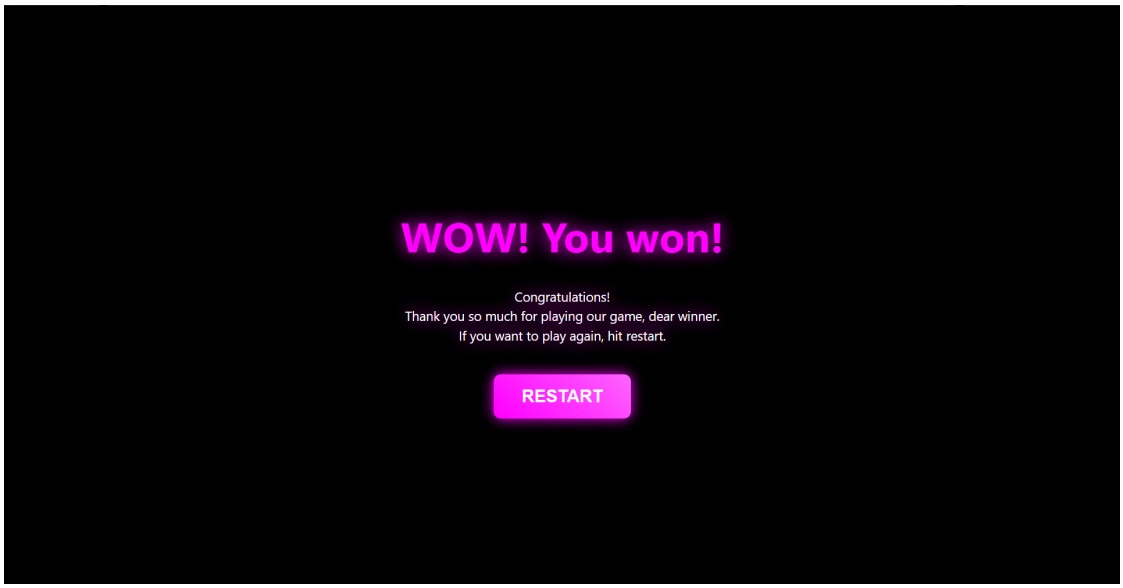

GAME INSTRUCTIONS
Grow the snake as much as possible.
Use arrow keys to move.
Get 20 points to win!
Do NOT hit the sides or the snake itself.
Your microphone controls targets.
The louder, the more targets appear.
Click and drag your mouse around to rotate the camera.
PLAY


Score: 1


Score: 1
Game Over!
We told you not to touch the sides!
To play again, hit restart.
RESTART


WOW! You won!
Congratulations!
Thank you so much for playing our game, dear winner.
If you want to play again, hit restart.
RESTART

Bibliography

1. Weiquan Mao & Jiyao Yuan. "*3D Snake Game*". Stanford Education. 2019. https://stanford.edu/class/ee267/Spring2019/report_yuan_mao.pdf

2.Mina Pêcheux. "*Making a 3D web runner game #1: Creating a basic 3D scene with Three.js* ". Medium. August 23, 2021. https://medium.com/geekculture/making-a-3d-web-runner-game-1-creating-a-basic-3d-scene-with-three-js-56b02635b5d2

3. Texture Labs. 2026. https://texturelabs.org/